

Chapter 2

Network Models

Objectives

- To discuss the OSI models and its architecture and to show the interface between the layers
- Briefly discuss the function of each layers
- Introduce TCP/IP protocol
- Similarities and comparison between OSI and TCP/IP model
- encapsulation

Layered tasks

- We use the concept of **layers** in our daily life
- As an example, let us consider 2 friends who communicate through postal mail.
- The process of sending a letter to a friend would be complex if there were no services available from the post office.

Process of sending a letter

- Sender, Receiver, and Carrier
- Hierarchy
 - Tasks must be done in order given in hierarchy
- Services
 - each layer at the sending site uses the services of the layer below it.

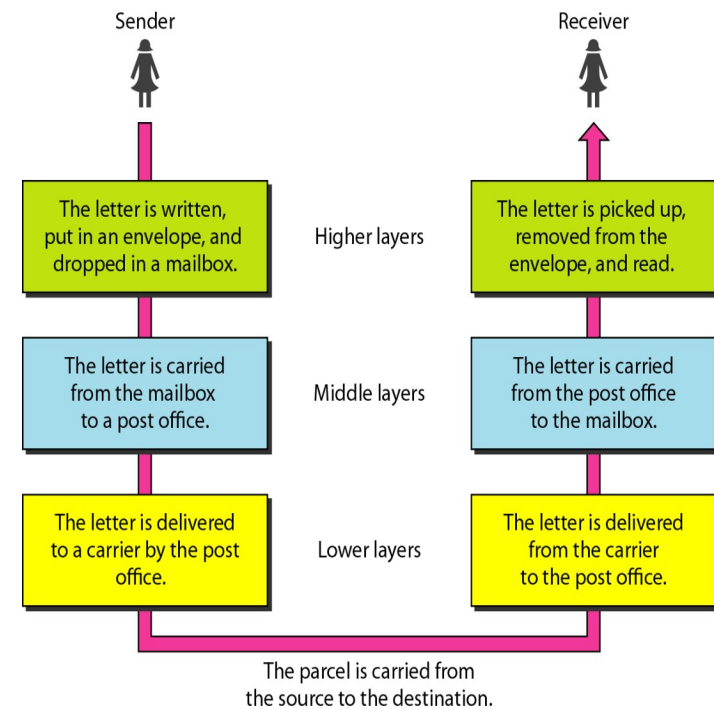
Tasks involved in sending a letter

- A network model is a layered architecture

- Task broken into subtasks
- Implemented separately in layers in stack
- Functions need in both systems
- Peer layers communicate

- Protocol

- A set of rules that governs data communication
- It represents an agreement between the communicating devices



Network Model

- Network model is a method of describing and analyzing data communication networks by breaking the entire set of communication process into a number of layers
- Each layer has a specific function

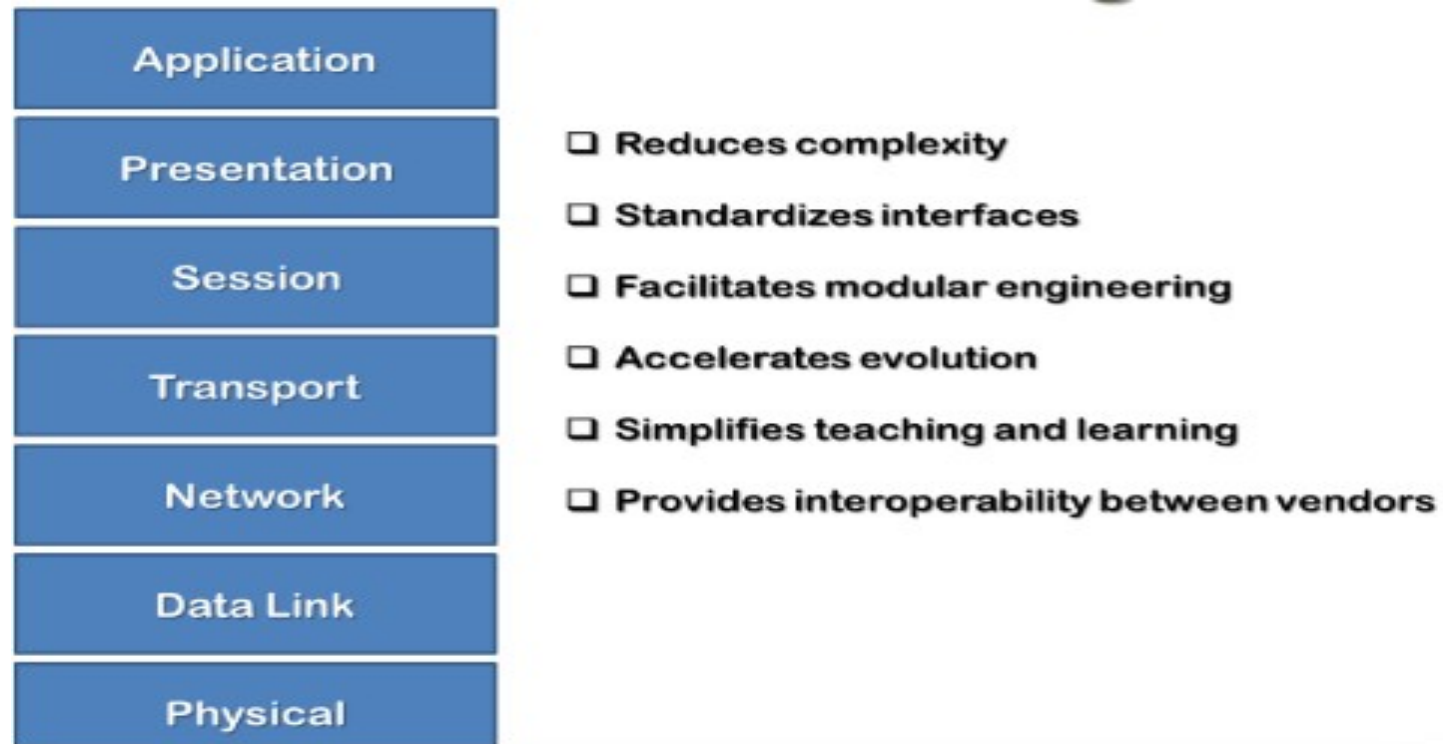
The OSI model

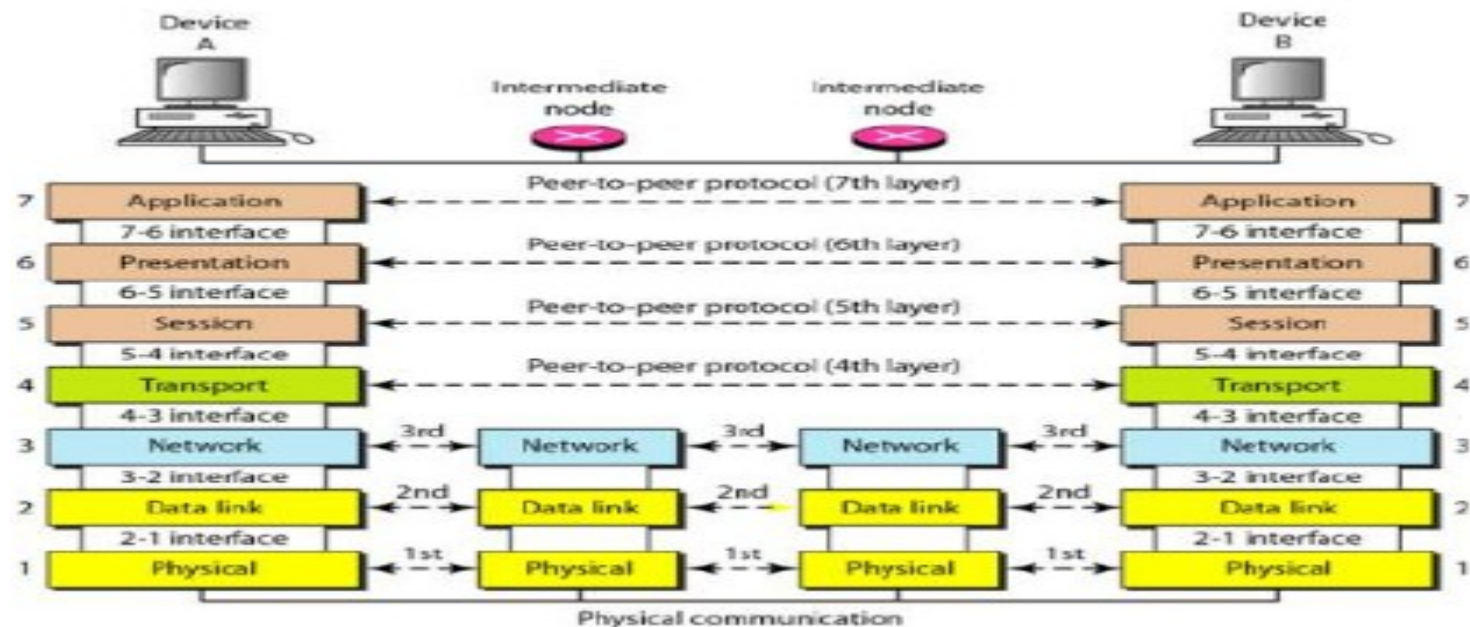
- The OSI model is based on a proposal developed by the International Standards Organization (ISO) as a first step toward international standardization of the protocols used in the various layers (Day and Zimmermann, 1983). It was revised in 1995 (Day, 1995).
- The model is called the ISO-OSI (Open Systems Interconnection) Reference Model because it deals with connecting open systems—that is, systems that are open for communication with other systems.

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- It is not a protocol: it is a model for understanding and designing a network architecture that is flexible and robust
- The OSI model is a layered framework for the design of network systems that allows communication between all types of computer systems.
- It consists of seven separate but related layers, each of which defines a part of the process of moving information across a network.

Why a layered Network Model





At the physical layer, communication is direct. At higher layers, communication must move down through the layers on Device A, then back up the layers on Device B. Each layer adds its own information to the layer above it, and passes the whole thing along.

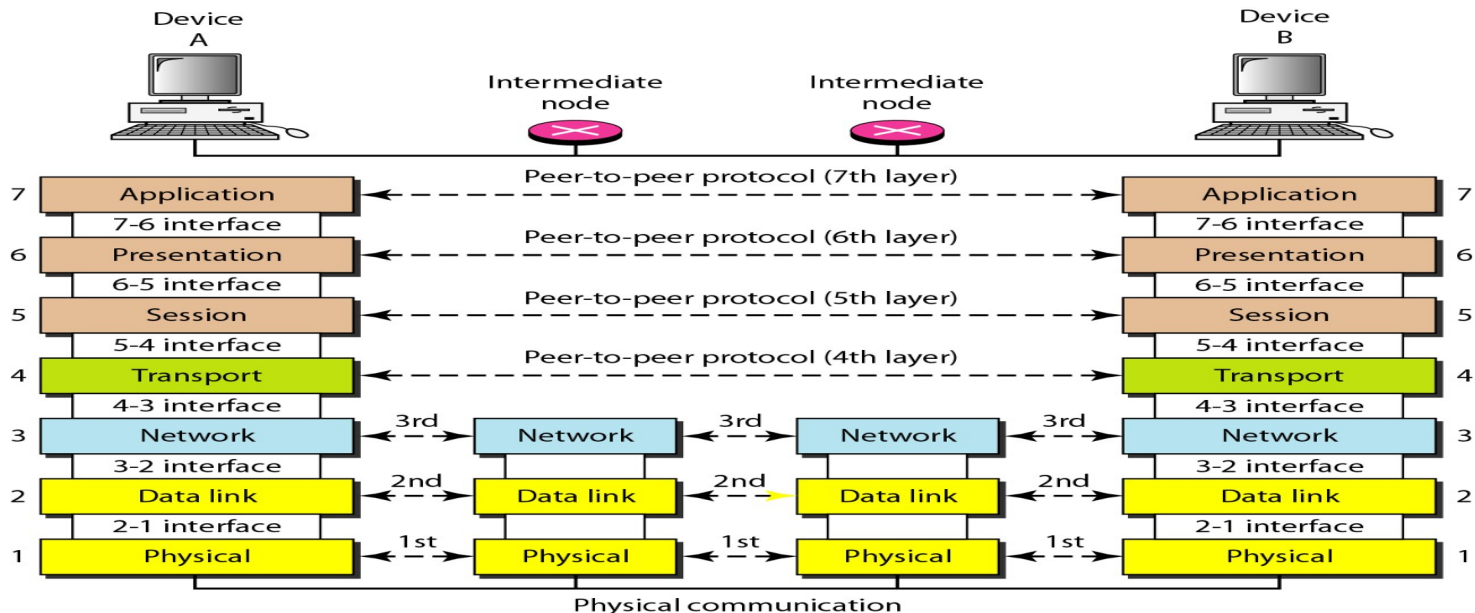
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Layer organization

- **Network support layers :layer 1,2,3**
 - Deals with physical aspect of moving data from one device to the other.
 - Ex: electrical specifications, physical connections, physical addressing,
- **User support layers :**
 - Allow interoperability among unrelated software systems.
- **Layer 4: the transport layer**
 - Links the two subgroups and ensures that what the lower layers have transmitted is in a form that upper layers can use.

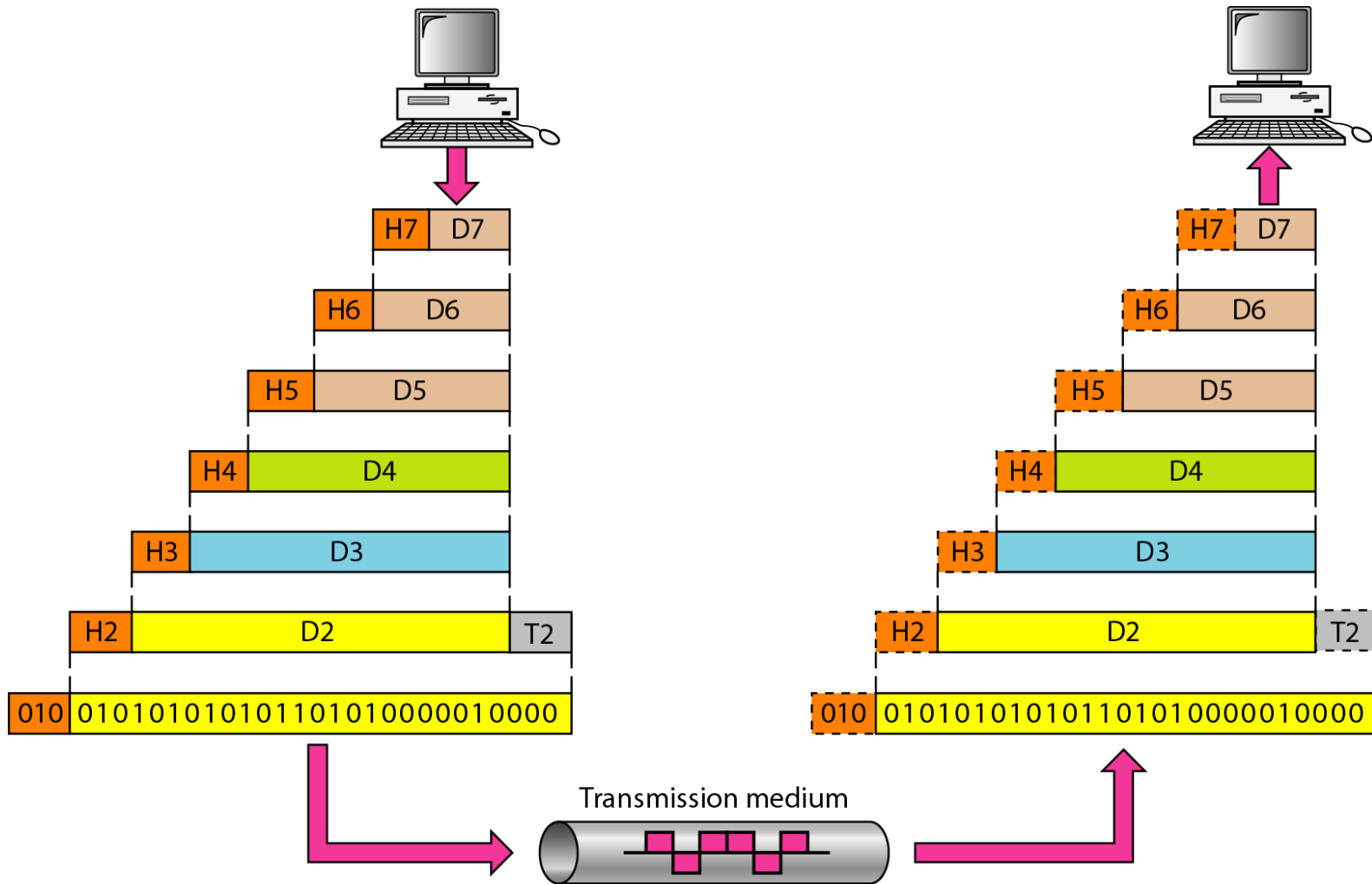
The interaction between layers in the OSI model

- Communication between peer layers is “virtual”
 - In reality, communications pass up and down the protocol stacks on both machines
 - As data gets passed from layer to layer, it’s divided into data units appropriate for the layer
 - **Protocol data units (PDUs)** are passed as a self-contained data structure from layer to layer
 - **Encapsulation** process adds “headers” to allow successful delivery of each layer’s payload
 - **Decapsulation** strips header information on way up
 - No layer can pass information directly to its peer counterpart except for the Physical layer



Encapsulation Process

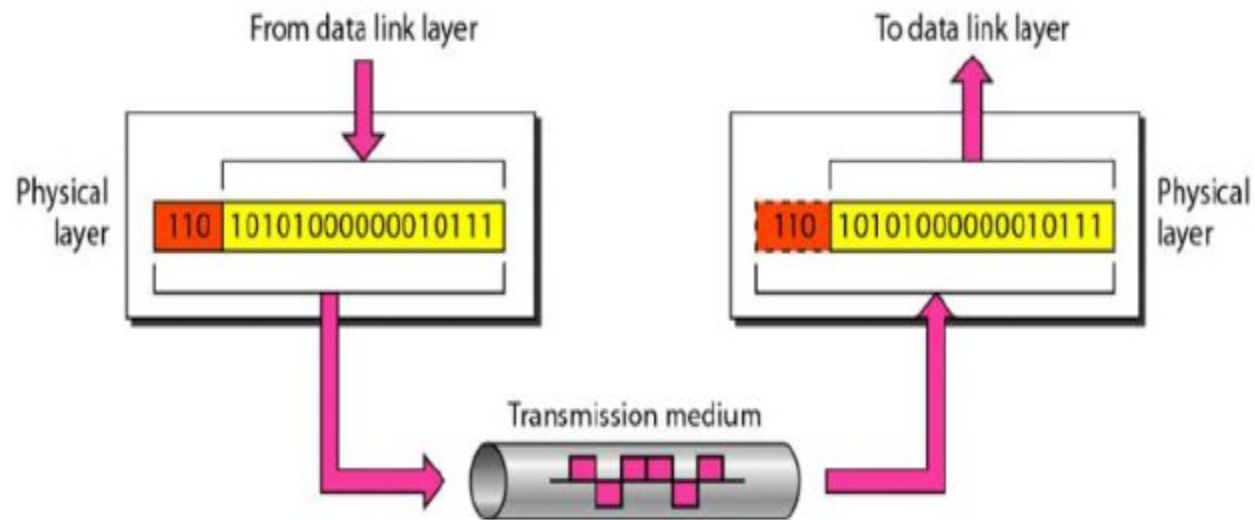
OSI Model			
	Layer	Data Unit	Function
User support layer	Application	DATA	Access to Network Resources
	Presentation		Data representation: translate compress and encryption
	Session		Control: the dialogues , Establishes, manages and terminates the connections between the local and remote application
Link	Transport	Segment	End-to-end connections and reliability
Network support layer	Network	Datagram/Packet	Path determination (Routing) and logical Addressing
	Data link	Frame	Hop to Hop delivery and Physical addressing
	Physical	Bit	Media, signal and binary transmission,...



Physical Layer

- The physical layer coordinates the functions required to carry a bit stream over a physical medium.
- It deals with the mechanical and electrical specifications of the interface and transmission medium.
- It also defines the procedures and functions that physical devices and interfaces have to perform for transmission to occur.
- Moves individual bits from one node to the next.

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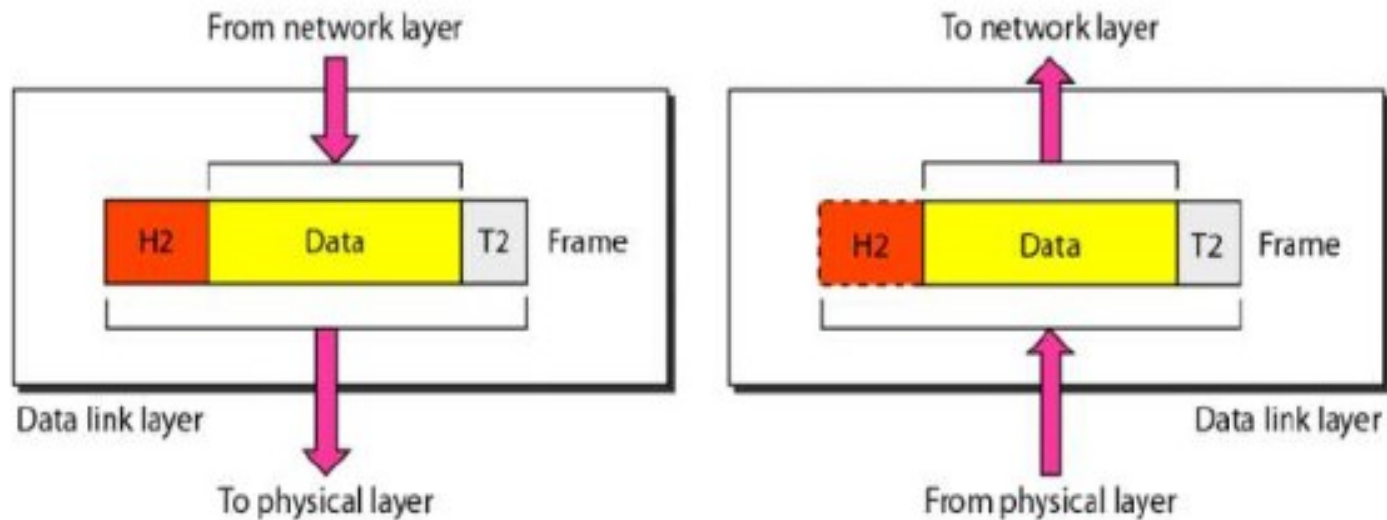
Other concerns

- Physical characteristics of interfaces and medium.
- **Representation of bits**
 - Physical Layer Data consists of a stream of bits (0's or 1's) with no interpretation.
 - Physical defines type of encoding (how 0's/1's are changed to signals) -- electrical or optical.
 - To be transmitted, bits must be encoded into signals. The physical layer defines the type of encoding
- **Data rate.**
 - Transmission rate: Number of bits sent per second.
 - Implies PHY defines the duration of a bit. which is how long it lasts.
- **Synchronization of bits.**
 - The sender and receiver clocks must be synchronized.

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- **Line Configuration.**
 - The Physical Layer is concerned with the connection of devices to the media. (Point-to-point vs multipoint.)
- **Physical topology**
- **Transmission mode**
 - Simplex, Half-duplex, full-duplex
- **Multiplexing**
- **Switching**

Data link Layer



- Transforms the Physical layer into a reliable link. Makes Physical layer appear error-free to the upper layer.
- Moves frames from one node to the next

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Framing

- Divides the stream of bits received from network layer into manageable data units called frames.

Physical addressing.

- If frame going to somewhere in same network:
 - Adds a header to the frame to define sender/receiver of the frame.
- If frame intended for outside the current network,
 - Receiver address is the one that connects this network to the next one.

Flow control.

- Try to avoid overwhelming the receiver
 - If $\text{rate}(\text{receiver data absorption}) < \text{rate}(\text{sender data produced})$

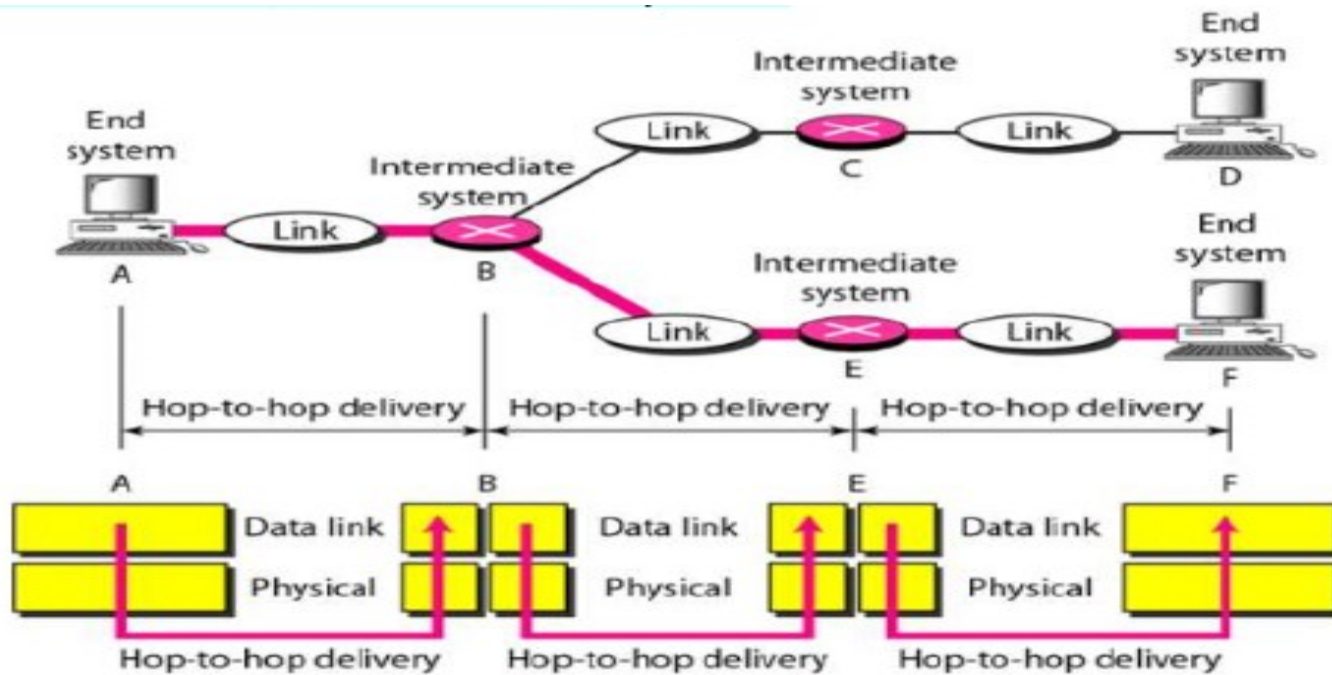
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Error Control.

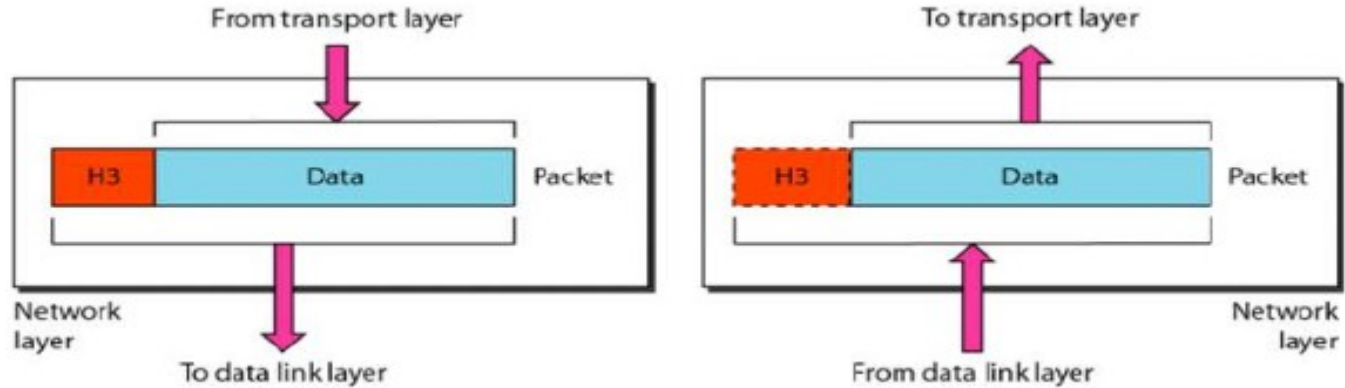
- Add mechanisms to detect and retransmit damaged or lost frames
- Typically achieved through trailer added to end of frame.
- Recognizes duplicate frames.

Access Control

- When two or more devices connected to the same link, data link layer determines which device has control over the link at any time.
- **Feedback**



Network Layer



- Responsible for source-to-destination delivery of packets, across multiple networks.
- Data Link can only deliver the packet within a network.

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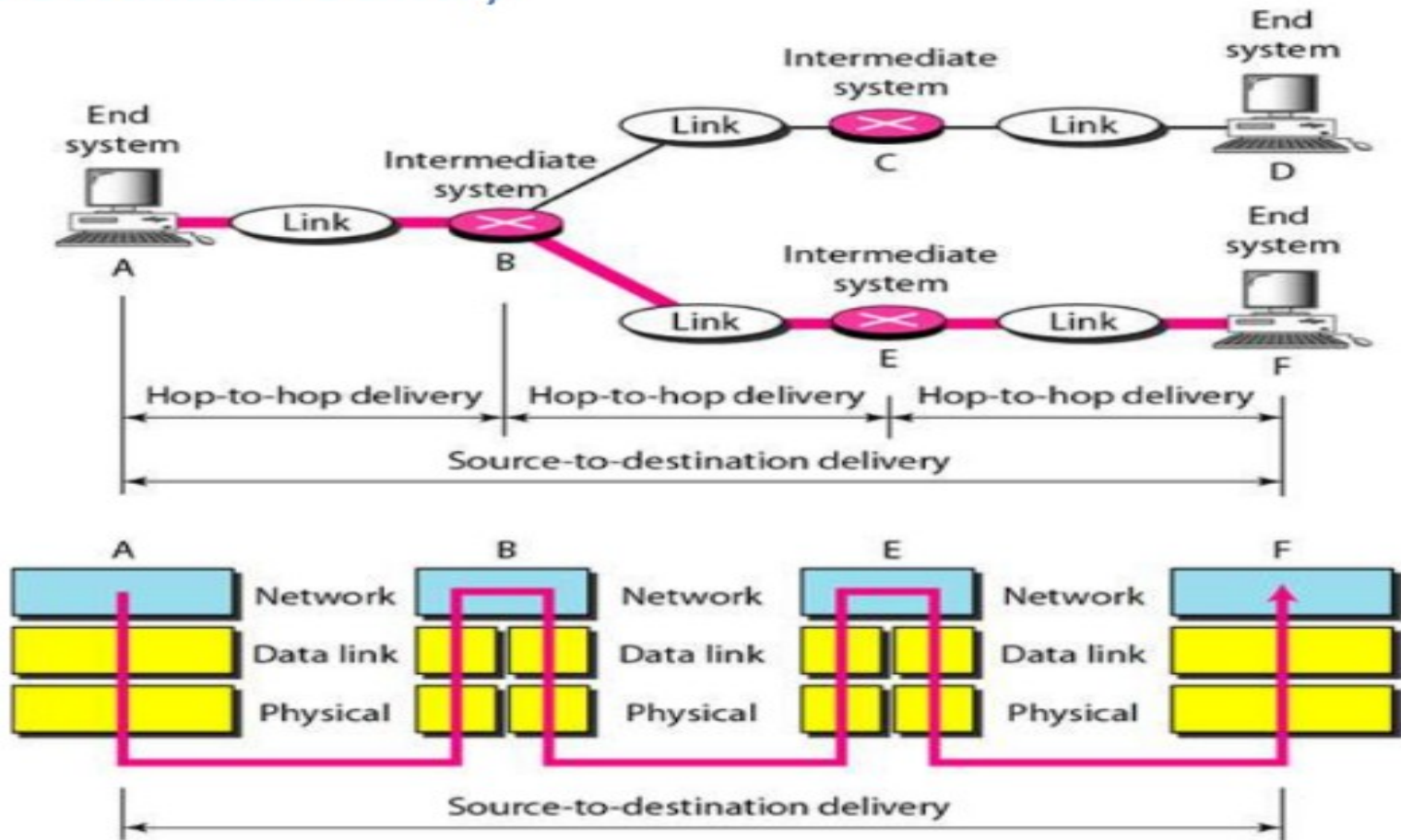
Logical Addressing

- If a packet passes the network boundary, need another addressing system to help distinguish the source and destination systems.
- Network layer adds a header to the packet coming from the upper layer that contains the logical address of the sender and receiver.

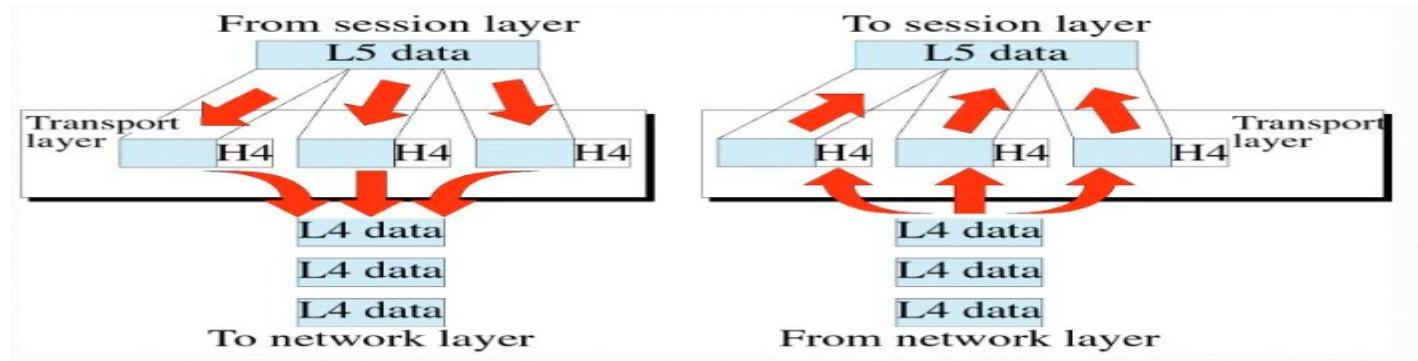
Routing

- When independent networks are connected to create internetworks, the routers/switches send packets to their final destination. The network layer does this.

Source to Destination Delivery



Transport Layer



- The network layer does not recognize any relationship between packets.
- The Transport layer ensures the whole message (group of packets) arrives in tact and in order.
- The transport layer also is responsible for delivery of a message from one process to another.

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Service-point addressing.

- "Computer addresses" have ports (22, 80, etc.).
- The Transport Layer Header defines a service-point address (port address).
- The transport layer make sure that the message is delivered to the correct process on the destination machine

Segmentation and re-assembly.

- message is divided into transmittable segments, each containing a sequence number. On receipt, assemble message in order, replace lost packets.

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Connection control.

- Can be connectionless or connection-oriented.
 - **Connectionless:** Treats each segment as an independent packet. the receiver doesn't acknowledges the receipt of a packet.
 - **Connection-oriented:** Makes a connection with the transport layer at the destination machine first before delivering packets. After all data transferred, connection is terminated. the receiver acknowledges the receipt of a packet.

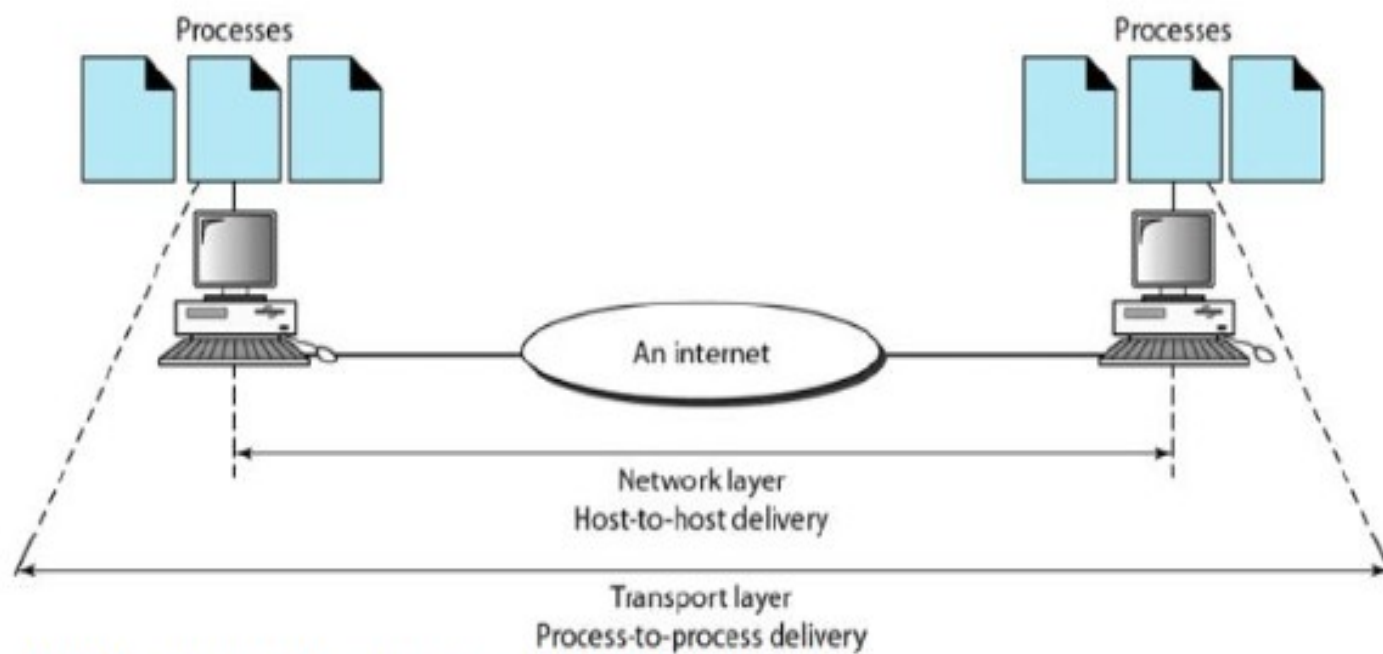
Flow control.

- Like the Data Link Layer, Transport is also responsible for flow control.
- Flow control at this layer is performed end-to-end rather than link-to-link.

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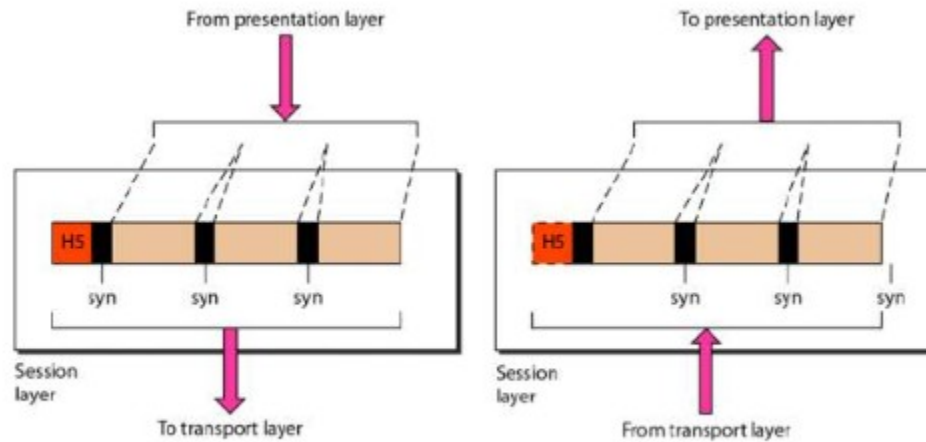
Error Control

- Performed process-to-process rather than across a single link.
- Sending transport layer makes sure entire message arrives at receiving transport layer
- without error.
- Error Correction typically achieved through retransmission.



Reliable process-to-process delivery of a message.

Session Layer



he session layer is the network **dialog controller**. It establishes, maintains, and synchronizes the interaction among communicating systems which is called a **session**.

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- **Establishment, maintaining and ending a secession:**
 - Send SYN packet– establishment request
 - Receive ACK and SYN--established
 - To end –sender sends ACK

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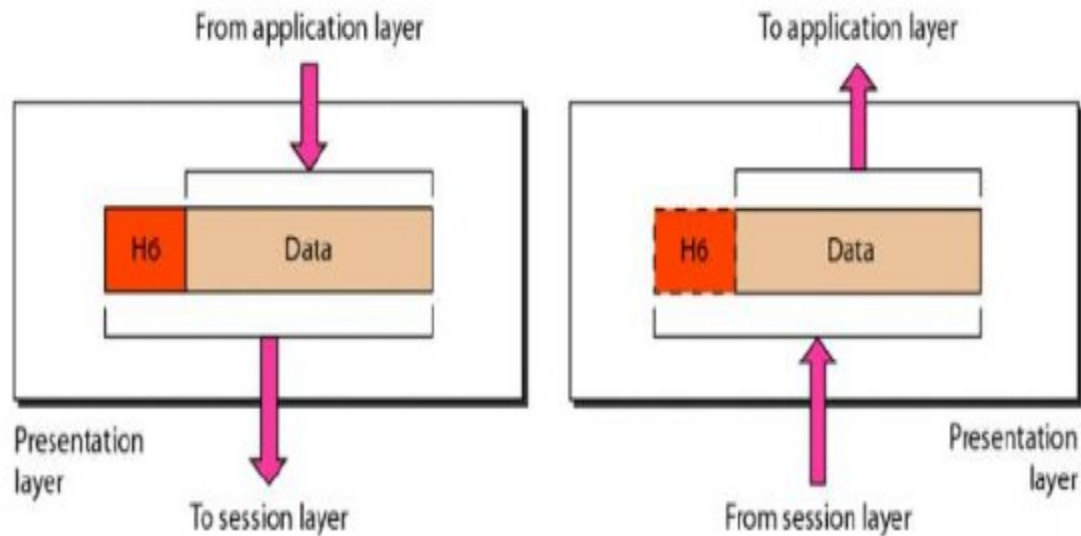
Dialog Control:

- Allow two systems to enter into a dialog. Allows communication between two processes to be half-duplex or full-duplex.

Synchronization:

- Session layer allows a process to add checkpoints, or synchronization points to a stream of data.
- Example: If sending a 2000 page file, add checkpoints every 100 pages to make sure that each 100-page unit is received and acknowledged independently. If a single 100-page section transmission fails, no need to resend the whole file, just that section.

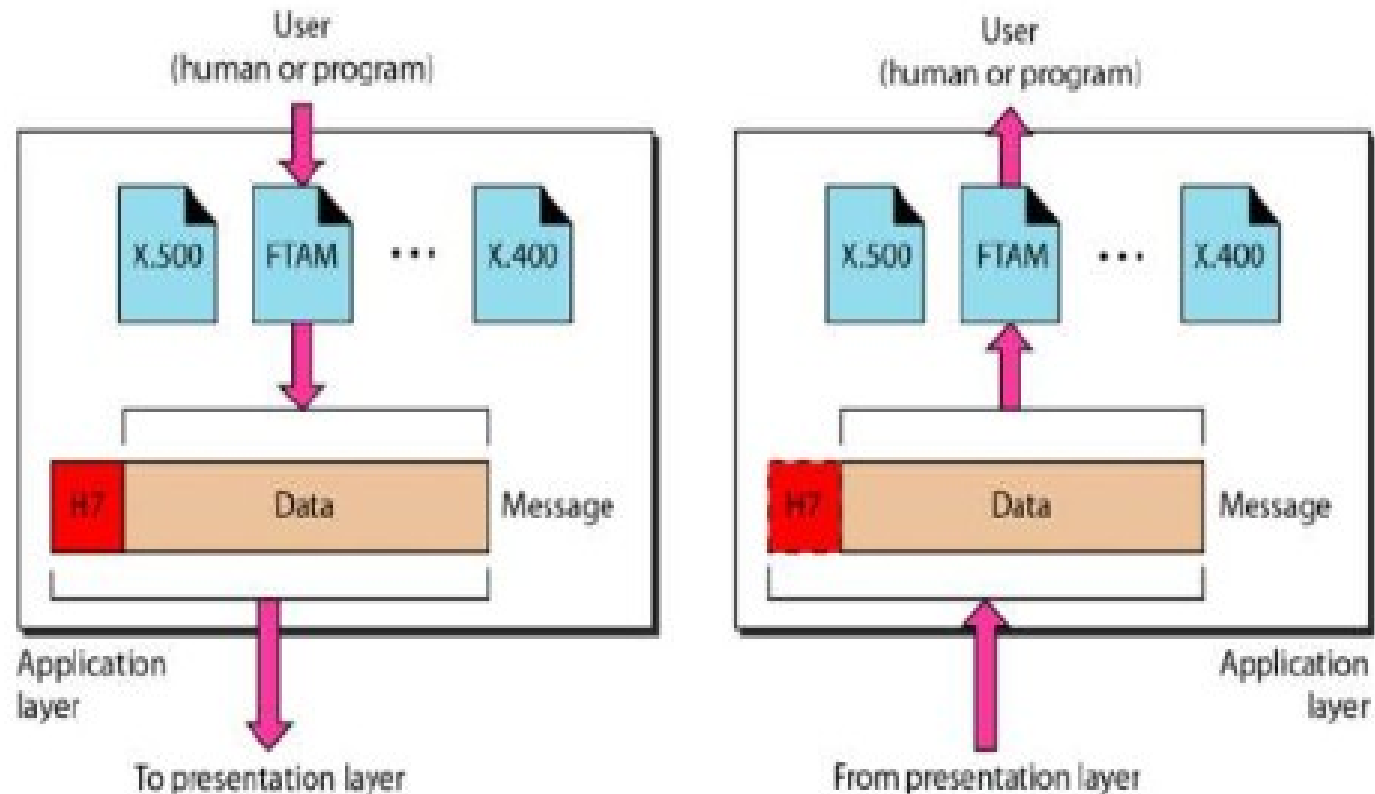
Presentation Layer



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- **Translation**
 - Changes sender-dependent format into common format. /vise-versa.
 - Example: Character encodings.
- **Encryption**--for security and privacy purpose
- **Compression**—reduce the number of bits contained in the information

The application layer



The application layer

- The application layer enables the user, whether human or software, to access the network.
- It provides user interfaces and support for services such as electronic mail, remote file access and transfer, shared database management, and other types of distributed information services.

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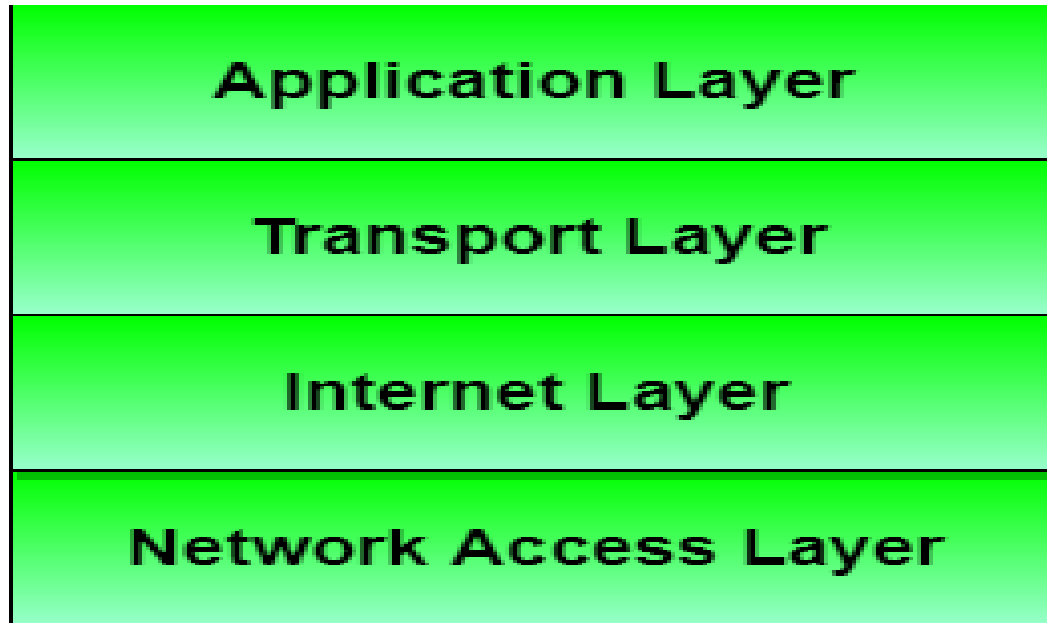
Specific services provided by the application layer include the following:

- Network virtual terminal: this application allows a user to access files on a remote host.
- File transfer, access, and management
- Mail services: this application provides various e-mail services
- Directory services: this application provides the distributed database sources and access for global information about various objects and services

TCP/IP Model

- The TCPIIP protocol suite was developed prior to the OSI model.
- TCP/IP forms the base of present day internet
- TCP and IP are two protocols of this model.
- This model was initially used by ARPNET
- The original TCP/IP protocol suite was defined as having four layers: host-to-network, internet, transport, and application.

TCP/IP protocol Suite



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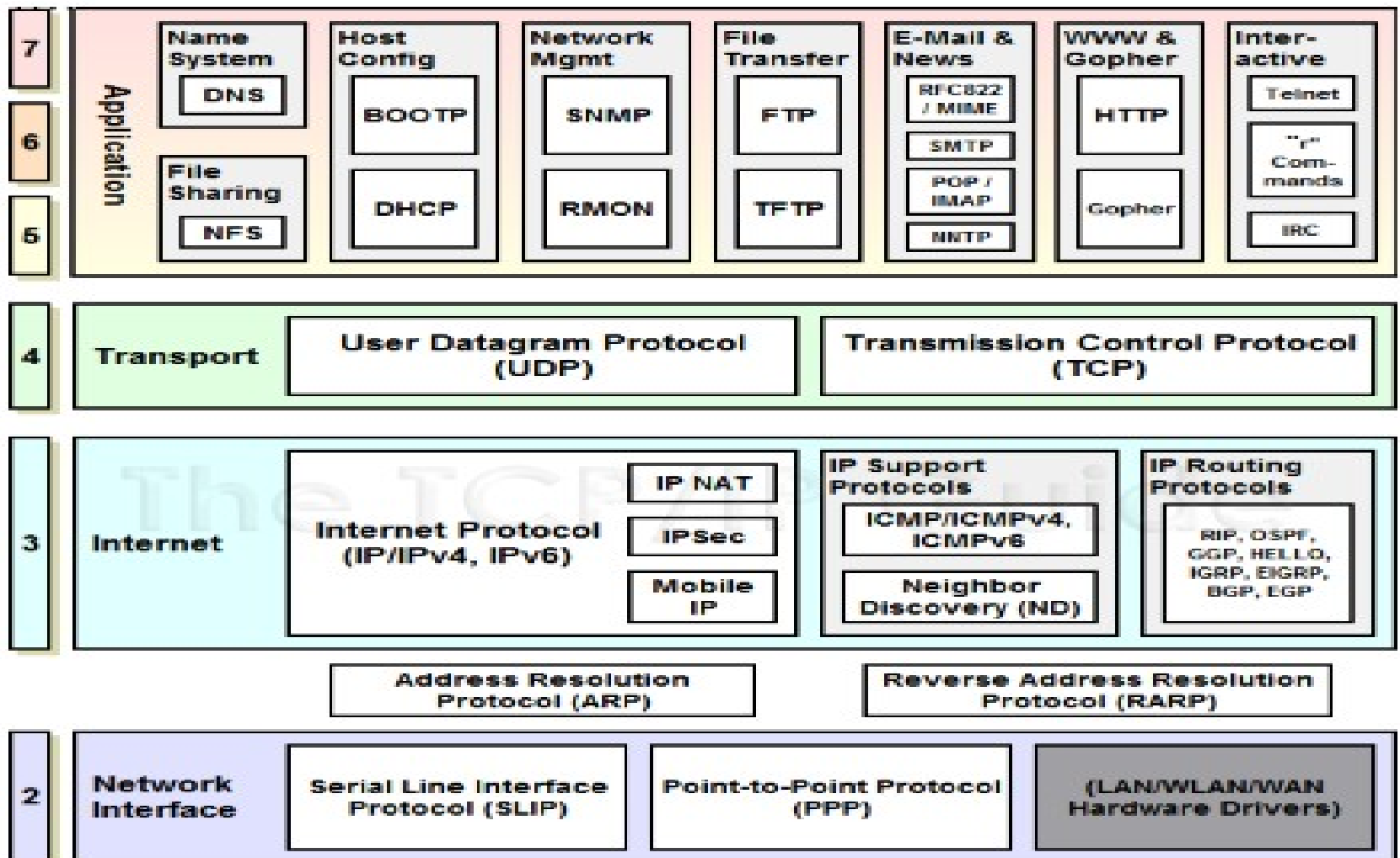
TCP/IP is compared to OSI, we can say that the

- **host-to-network layer** is equivalent to the combination of the physical and data link layers.
- **internet** layer is equivalent to the network layer, and the
- **transport layer** in TCP/IP taking care of part of the duties of the session layer.
- **application layer** is roughly doing the job of the session, presentation, and application layers

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- TCP/IP is a hierarchical protocol made up of interactive modules, each of which provides a specific functionality; however, the modules are not necessarily interdependent.
- Whereas the OSI model specifies which functions belong to each of its layers,
- the layers of the TCP/IP protocol suite contain relatively independent protocols that can be mixed and matched depending on the needs of the system.
- The term hierarchical means that each upper-level protocol is supported by one or more lower-level protocols.

TCP/IP Protocols



Application Layer

- This layer is comparable to the application, presentation, and session layers of the OSI model all combined into one.
- It provides a way for applications to have access to networked services.
- This layer also contains the high level protocols. The main issue with this layer is the ability to use both TCP and UDP protocols.

Transport Layer

- It is similar in functionality with the OSI model transport layer.
- Provides connectionless and connection oriented services
 - Connection oriented-- TCP
 - Connection less --UDP

Internet Layer

- Similar to the Network layer of the OSI model.
- The main purpose of this layer deliver IP packet to their destination.
- The main protocol used at this layer is IP.
While ICMP(used by popular 'ping' command) and IGMP are also used at this layer.

Host to Network Layer

- This layer is also known as network interface layer
- The purpose of this layer is to connect the host to the network.

Similarities of OSI and TCP/IP

- Both are based on the concept of stack of independent protocols
- Functionality of layers is roughly similar
- Up to transport –network oriented
- Above--- user oriented

Difference between OSI and TCP/IP

- OSI model has seven layers and TCP/IP has four layers.
- OSI model provides clear distinction between Services, Interfaces, and Protocols.
- TCP/IP doesn't provides clear distinction between Services, Interfaces, and Protocols.
- The OSI model transport layer is connection oriented.
- The TCP/IP transport layer is both connection oriented and connectionless.

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- The protocols do not fit well into the OSI model
- Protocols fit well in the TCP/IP model
- The minimum size of the OSI header is 5 byte in TCP/IP it is 20 bytes

Communication Protocols

Definition

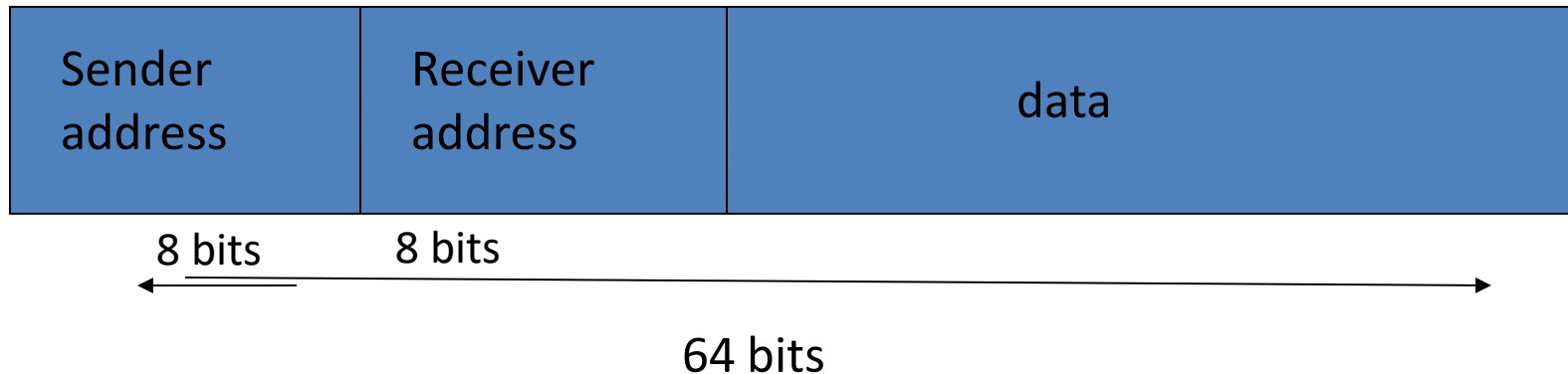
- Protocol is a set of rules that govern all aspect of data communication between computers on a network.
- These rules include guidelines that regulate the following characteristics of a network: access method, allowed physical topologies, types of cabling, and speed of data transfer.
- A protocol defines what, how, when it communicated.
- The key elements of a protocol are syntax, semantics and timing.
- Protocols are to computers what **language** is to humans. Since this article is in English, to understand it you must be able to read English. Similarly, for two devices on a network to successfully communicate, they **must both understand** the same **protocols**.

Elements of protocol

i) Syntax

The structure or format of the data.

Eg. A simple protocol;



ii) Semantics

- Refers to the meaning of each section of bits.
- how is a particular pattern to be interpreted, and what action is to be taken based on that interpretation.

Eg. Does an address identify the route to be taken or the final of the message?

iii) Timing

Refers to two characteristics:

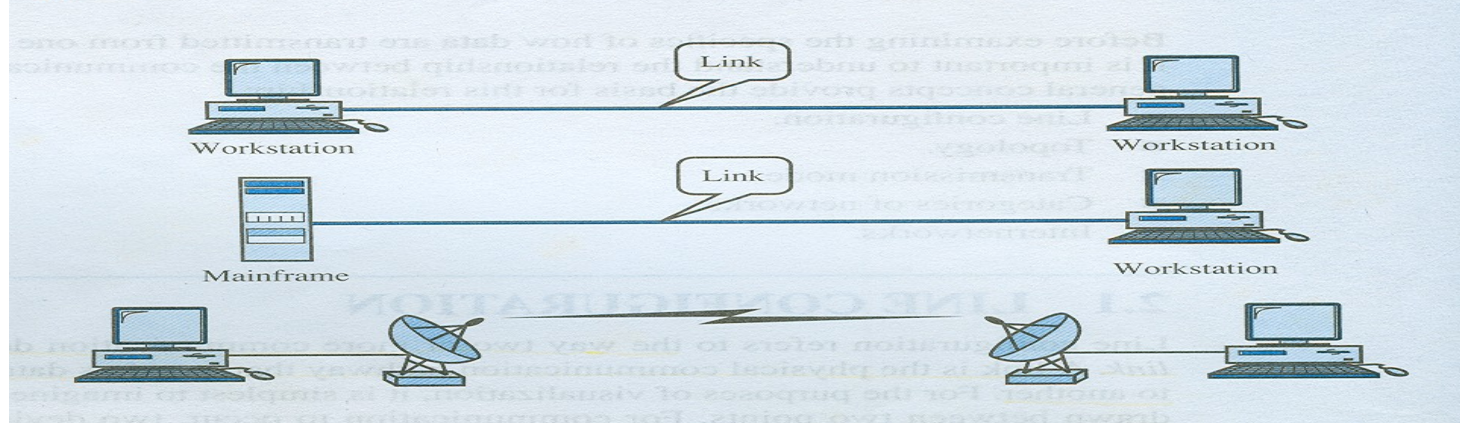
- a. When data to be sent
- b. How fast it can be sent

Eg. If a sender produces data at 100 Mbps but the receiver can process data at only 1 Mbps, the transmission will overload the receiver and data will be largely lost.

Characteristics of a protocol

- Direct/indirect
- Monolithic or structure
- Symmetric or Asymmetric
- Standard Propriety and De facto

Figure 2.2 *Point-to-point line configuration*



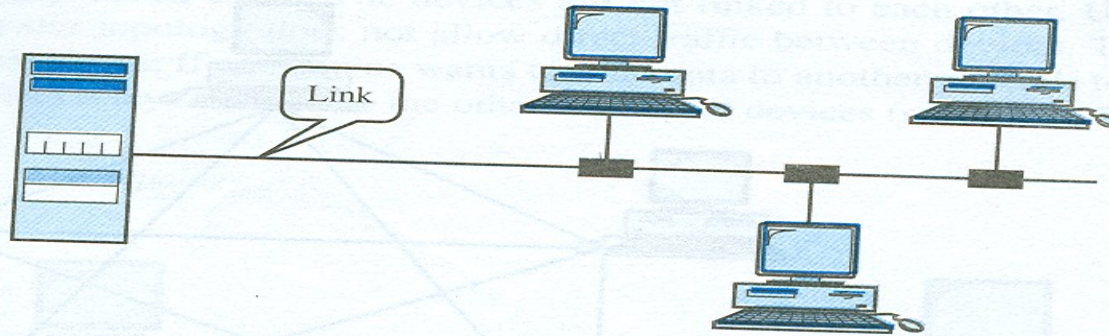
a) Direct / indirect

- communication between two entities maybe direct or indirect.

i) point-to-point link

- connection provides a dedicated link between two devices
- the entities in these systems may communicate directly that is data and control information pass directly between entities with no intervening active agent.

Figure 2.3 *Multipoint line configuration*



ii) multipoint link/

- more than two devices can share a single link
- Entities depend on other entities for data exchange .
- Entities may be connected over local network (Ethernet) or may belong to different networks (internet) , Switched networks
- The entities must be concerned with the issue of access control and making the protocol more complex.

Monolithic/structured protocol

- Monolithic protocol
 - All protocol logic (for every layer) is combined into one unit .
 - Problem in changing any aspect of the protocol (such as virtual circuit request)
- Structured protocol
 - Set of protocols with hierarchical/layered structure .
Clearly shows the distinction between different layers of logic
- Communications architecture
 - Hardware/software used to implement communications with structured protocol

c) **Symmetric / asymmetric**

- **Symmetric** is the most use in protocol and involve communication between peer entities.
- **Asymmetry** may be dictated by the logic of an exchange (eg; client and a server process) the desire to keep one of the entities or systems as simple as possible.

d) Standard / nonstandard

Standard protocol : Commonly accepted protocols that have been agreed on by a standards body

Nonstandard protocol : Built for specific communications situation

Common protocol used

Protocol	Acronym	Remarks
Point To Point	PPP	Used to manage network communication over a modem
Transfer/Transmission Control Protocol	TCP / IP	Backbone protocol. The most widely used protocol.
Internetwork package exchange	IPX	Standard protocol for Novell NOS
NetBIOS extended user interface	NetBEUI	Microsoft protocol that doesn't support routing to other network. Running only Windows-based clients.
File transfer Protocol	FTP	used to send and received file from a remote host
Simple mail Transfer protocol	SMTP	Used to send Email over a network
Hyper text transfer protocol	HTTP	Used for Internet to send document that encoded in HTML
Apple Talk	Apple Talk	Protocol suite to network Macintosh computer and a peer-to-peer network protocol
OSI Model	OSI Layers	A way of illustrating how information functions travels through network of its 7 layers.

Encapsulation

- Data as well as control information in each pdu
- Protocols and Architecture
- Control information is divided into the following categories:
 - Address of the sender and receiver
 - Error detection code or frame check sequence
 - Protocol control for other protocol functions

Segmentation and reassembly

- Segment the data stream into small bounded size blocks or pdus
- Reasons for segmentation
 - Communications network may accept data blocks only up to a certain size (53 octets for atm, 1526 octets for Ethernet)
 - Efficient error control with smaller pdu size; fewer bits retransmitted in the event of failure
 - Better access to shared transmission facilities, with shorter delay; nobody can monopolize the network ·
 - Smaller buffers at receiver stations
 - Can pause transfer for checkpoint and recovery

Disadvantages with segmentation

- Larger overhead with smaller pdu size ·
- More interrupts as pdus announce their arrival
- More time spent to process smaller pdus
- * Segmented data is reassembled into messages appropriate for application level

Connection control

- Connectionless data transfer .
 - Each pdu is independent of other pdus
- Connection-oriented data transfer .
 - Used if stations are to be connected for long time or protocol details are to be worked out dynamically .
 - Also known as logical association, or virtual circuit, with three phases
 1. Establish connection
 2. Transfer data
 3. Terminate connection

Establish connection

- One station issues a connection request to the other, with or without involving a central authority
- Receiver may accept or reject the connection .
- Request may include negotiating syntax, semantics, and timing of protocol
- Protocol may have some options to be negotiated at connection time, such as pdu size

- Transfer data
 - Exchange data and control information (flow control, error control) .
 - Data flows in one direction while acknowledgements flow in the other
- Terminate connection
 - Either side may terminate connection by sending a request .
 - Connection may be terminated by a central authority

Sequencing

- pdus are sequentially numbered as they are sent .
- Each side keeps track of outgoing numbers (generated locally), and incoming numbers (generated by other host)

Ordered delivery

- pdus may not arrive in order in which they are sent
- Connection-oriented protocols require the pdu order to be maintained Protocols and Architecture 12
- Number the pdus sequentially as they are generated
- Problem if sequence numbers repeat after overflow
- Preferable to have the maximum number pdu to be twice the maximum number of outstanding pdus

Flow control

- Function of receiving entity to limit the amount of data sent by transmitter
- Stop-and-wait · Each pdu must be acknowledged before next one can be sent
- Efficiency requires the transmission of a fixed number of pdus without acknowledgement
- Implemented in several protocols

Error control

- Guard against loss or damage of data and control information
- Implemented as error detection and retransmission
- Detection of error by receiver makes him discard the pdu
- No acknowledgement makes the sender retransmit the pdu
- With error correction code, the receiver may be able to correct the error at destination
- Implemented in several protocols

Addressing

- **Addressing level** · Level in the communications architecture at which an entity is named · Network-level address or ip address used to route a pdu through network (also called Network Service Access Point or nsap in osi terminology) · Upon arrival at destination, the pdu must be routed to port or Service Access Point (sap) for the application
- **Addressing scope** · Global address Global nonambiguity One system to one address but possible to have more than one address for the system Global applicability Any system can be identified from anywhere · A port may not be unique in the network unless attached to a system (think of smtp port on every system in a network)

Connection identifiers

- Useful for connection-oriented data transfer (virtual circuit) but meaningless for connectionless data transfer (datagram) ·
- Identified by a connection name during the data transfer phase ·
- Advantages
 - **Reduced overhead** Data packets can contain just the circuit number after a virtual circuit is established
 - **Routing** Allows the setting up of a fixed route
 - **Multiplexing** More than one connection between entities; incoming pdu can be identified by connection identifier
 - **Use of state information** State information related to the connection; enables flow control and error control using sequence numbers

Addressing mode

- Individual or unicast address – reference to a single system or port ·
- Multicast or broadcast address

Multiplexing

- Combining several signals for transmission on some shared medium Protocols and Architecture
- Possible to have multiple virtual circuits terminating into a single end system
- Can also be accomplished via port names
- Upward multiplexing · Multiple higher-level connections are multiplexed on a single lower-level connection · Connecting your pc to isp for multiple applications, including web, email, telnet, ftp, ...
- Downward multiplexing · Split a single higher-level connection over a number of lower-level connections · Useful for reliability, performance, or efficiency

Transmission services

- Additional services, such as priority, quality of service, and security